

Inverting mixed symbols without commuting matrix factors

The inversion problem addressed here is pointwise in phase space. Its output is a symbol with quantitative derivative estimates, so it can subsequently enter an operator construction. No composition or boundedness theorem for pseudodifferential operators is needed for the inversion itself.

From symbol estimates to operators on every Sobolev scale supplies the earlier symbol-space context. The calculation below uses multivariable differentiation, the product rule, matrix norms and smooth cutoffs; every mixed estimate needed here is proved directly.

1. The two frequency weights

Fix integers $n, N \geq 1$. Write

$$x = (x_1, \dots, x_n) \in \mathbb{R}^n, \quad \xi = (\xi', \xi_n) \in \mathbb{R}^{n-1} \times \mathbb{R}. \quad (\text{MSI1})$$

The matrix space is $M_N(\mathbb{C})$, equipped with the operator norm induced by the Euclidean norm on $\mathbb{C}^N$. Thus $\|UV\| \leq \|U\|\|V\|$, $\|I_N\| = 1$, and scalar multiplication has its usual absolute-value norm. For $n = 1$, the vector ξ' has no coordinates, its norm is zero, and every multiindex in its coordinates has length zero.

For real numbers p, q and multiindices $\alpha, \beta \in \mathbb{N}_0^n$, put $\alpha' = (\alpha_1, \dots, \alpha_{n-1})$ and

$$W_{p,q;\alpha}(\xi) = (1 + |\xi|)^{p-\alpha_n} (1 + |\xi'|)^{q-|\alpha'|}. \quad (\text{MSI2})$$

The original weights $1 + |\xi|$ and $1 + |\xi'|$ are retained throughout. They are positive numerical weights; differentiability of these weights at the origin is neither assumed nor used.

A smooth function $a : \mathbb{R}_x^n \times \mathbb{R}_\xi^n \rightarrow M_N(\mathbb{C})$ belongs to $S^{p,q}$ when every number

$$\|a\|_{p,q;\beta,\alpha} = \sup_{x,\xi} \frac{\|\partial_x^\beta \partial_\xi^\alpha a(x, \xi)\|}{W_{p,q;\alpha}(\xi)} \quad (\text{MSI3})$$

is finite. In particular the constants are uniform in all $x \in \mathbb{R}^n$; there is no implicit compact restriction on x . The convention $D_j = -i\partial_j$, if used in an operator realization, gives the same seminorms, because each derivative of fixed multiindex differs by a scalar of modulus one. We use ordinary derivatives in the identities below so that every sign is explicit.

The product of two smooth matrix symbols satisfies the exact Leibniz identity

$$\partial_x^\beta \partial_\xi^\alpha (uv) = \sum_{\substack{\delta \leq \beta \\ \varepsilon \leq \alpha}} \binom{\beta}{\delta} \binom{\alpha}{\varepsilon} (\partial_x^\delta \partial_\xi^\varepsilon u) (\partial_x^{\beta-\delta} \partial_\xi^{\alpha-\varepsilon} v). \quad (\text{MSI4})$$

No factors on the right are interchanged. To prove this identity, apply the ordinary one-coordinate product rule repeatedly; choosing which of the β_j derivatives acts on the left factor gives $\binom{\beta_j}{\delta_j}$ choices, and the frequency derivatives give the second collection of binomial factors. Multiplying these independent counts gives (MSI4). Since

$$W_{p,q;\varepsilon} W_{r,t;\alpha-\varepsilon} = W_{p+r,q+t;\alpha}, \quad (\text{MSI5})$$

submultiplicativity and (MSI4) prove that $uv \in S^{p+r, q+t}$ whenever $u \in S^{p, q}$ and $v \in S^{r, t}$. More precisely, its (β, α) seminorm is bounded by the sum in (MSI4) with each differentiated factor replaced by its corresponding seminorm. This argument also applies on a subset of phase space whenever all derivatives in the displayed estimates exist in a neighborhood of each of its points.

Exact identity map to bracket-weight conventions

Some statements use $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$ and $\langle \xi' \rangle = (1 + |\xi'|^2)^{1/2}$. We retain the original weights and prove their precise correspondence with these additional weights. For every $s \geq 0$,

$$1 + s^2 \leq (1 + s)^2 \leq 2(1 + s^2),$$

because the first difference is $2s \geq 0$ and the second difference is $(s - 1)^2 \geq 0$. Thus the two positive ratios

$$\frac{1 + |\xi|}{\langle \xi \rangle}, \quad \frac{1 + |\xi'|}{\langle \xi' \rangle} \quad \text{belong to } [1, \sqrt{2}]. \quad (\text{MSI5a})$$

When $n = 1$, the second ratio is exactly one. For a real number t , put $t_+ = \max(t, 0)$, $t_- = \max(-t, 0)$. Raising a number in $[1, \sqrt{2}]$ to t gives a number in $[2^{-t_-/2}, 2^{t_+/2}]$; for $t < 0$ the inequality reverses on taking the reciprocal, which proves this assertion in that case as well.

Define the second, unchanged-derivative weight

$$\widetilde{W}_{p, q; \alpha}(\xi) = \langle \xi \rangle^{p - \alpha_n} \langle \xi' \rangle^{q - |\alpha'|}, \quad u = p - \alpha_n, \quad v = q - |\alpha'|. \quad (\text{MSI5b})$$

Multiplying the two ratio bounds gives, for these exact real exponents,

$$2^{-(u_- + v_-)/2} \widetilde{W}_{p, q; \alpha} \leq W_{p, q; \alpha} \leq 2^{(u_+ + v_+)/2} \widetilde{W}_{p, q; \alpha}. \quad (\text{MSI5c})$$

Let $\widetilde{\|f\|}_{p, q; \beta, \alpha}$ denote (MSI3) with only its denominator changed to $\widetilde{W}$. Division by the positive bounds in (MSI5c), followed by the same supremum over all x, ξ , proves

$$\begin{aligned} \|f\|_{p, q; \beta, \alpha} &\leq 2^{(u_- + v_-)/2} \widetilde{\|f\|}_{p, q; \beta, \alpha}, \\ \widetilde{\|f\|}_{p, q; \beta, \alpha} &\leq 2^{(u_+ + v_+)/2} \|f\|_{p, q; \beta, \alpha}. \end{aligned} \quad (\text{MSI5d})$$

Therefore the identity on the actual smooth matrix functions is a continuous linear bijection between the two symbol spaces with their families of seminorms. The two inequalities prove continuity in both directions. It commutes with every derivative, preserves matrix multiplication in its original order, and preserves the actual cutoff inverse b . These are literal identities on the functions, not changes of phase-space coordinates or of the operator convention.

The ellipticity constants also have definite transformations. Write c_R, C_R for scalar lower-bound and inverse-norm constants using $R = 1 + |\xi|$, and $c_{\langle \cdot \rangle}, C_{\langle \cdot \rangle}$ for constants using $\langle \xi \rangle$. The following choices are valid on exactly the same set:

$$\begin{aligned} |a| \geq c_R R^m &\implies c_{\langle \cdot \rangle} = c_R 2^{-m_-/2}, \\ |a| \geq c_{\langle \cdot \rangle} \langle \xi \rangle^m &\implies c_R = c_{\langle \cdot \rangle} 2^{-m_+/2}, \\ \|a^{-1}\| \leq C_R R^{-m} &\implies C_{\langle \cdot \rangle} = C_R 2^{m_-/2}, \\ \|a^{-1}\| \leq C_{\langle \cdot \rangle} \langle \xi \rangle^{-m} &\implies C_R = C_{\langle \cdot \rangle} 2^{m_+/2}. \end{aligned} \quad (\text{MSI5e})$$

For example the first implication follows from $R^m \geq 2^{-m_-/2} \langle \xi \rangle^m$; the second follows from $\langle \xi \rangle^m \geq 2^{-m_+/2} R^m$. The last two apply the corresponding upper bounds to exponent $-m$. This proves all four implications, including negative and zero orders. Together with (MSI5d), it transfers every hypothesis and conclusion of the inverse theorem in both directions with explicit constants, retaining both weight presentations.

2. An exact finite formula for inverse derivatives

Use a combined multiindex $\mu = (\beta, \alpha) \in \mathbb{N}_0^{2n}$, and write $\partial^\mu = \partial_x^\beta \partial_\xi^\alpha$. Inequalities and sums of combined multiindices are componentwise; $\mu! = \prod_{j=1}^{2n} \mu_j!$, and $0 < |\nu|$ means that ν is nonzero.

Let a be smooth on an open set $V \subset \mathbb{R}^{2n}$, with every $a(z)$ invertible. The inverse $r = a^{-1}$ is smooth: the cofactor formula expresses each entry of r as a polynomial in the entries of a divided by $\det a$, a nowhere-zero smooth function on V . For $\mu \neq 0$,

$$\partial^\mu r = \sum_{k=1}^{|\mu|} (-1)^k \sum_{\substack{\nu_1 + \dots + \nu_k = \mu \\ |\nu_i| > 0}} \frac{\mu!}{\nu_1! \dots \nu_k!} r(\partial^{\nu_1} a) r \dots (\partial^{\nu_k} a) r. \quad (\text{MSI6})$$

The inner sum is over ordered lists. This is essential: in general, two lists obtained by interchanging ν_i and ν_j produce different matrix products.

Here is a finite algebraic proof of (MSI6). Fix $z \in V$ and $L \geq |\mu|$. Work with polynomials in $2n$ formal commuting variables $h = (h_1, \dots, h_{2n})$, with matrix coefficients, and identify two polynomials when their difference contains only monomials of total degree greater than L . For any smooth matrix function f , form the finite polynomial

$$J_L f(h) = \sum_{|\eta| \leq L} \frac{\partial^\eta f(z)}{\eta!} h^\eta. \quad (\text{MSI7})$$

Formula (MSI4), coefficient by coefficient, proves $J_L(fg) = (J_L f)(J_L g)$ in this finite quotient algebra. Let $A = a(z)$, $R = A^{-1}$, and $H = J_L a - A$. Every monomial of H has positive degree. Consequently $(RH)^{L+1} = 0$ in the quotient. Direct multiplication, without commuting R and H , now gives

$$(A + H)^{-1} = \sum_{k=0}^L (-RH)^k R. \quad (\text{MSI8})$$

Indeed $A + H = A(I_N + RH)$, and $(I_N + RH) \sum_{k=0}^L (-RH)^k = I_N$, with the same identity in the reverse order. On the other hand, $J_L a J_L r = J_L r J_L a = I_N$, because $ar = ra = I_N$ as actual smooth functions. An inverse in any associative algebra is unique: if $XY = YX = I_N$ and $XZ = ZX = I_N$, then $Y = Y(XZ) = (YX)Z = Z$. Thus $J_L r$ equals (MSI8). The coefficient of h^μ in $(-RH)^k R$ is the ordered product sum in (MSI6), divided by $\mu!$. Comparing coefficients proves the formula at the arbitrary point z .

In particular, for a single coordinate z_j and for any two coordinates z_j, z_l ,

$$\partial_j r = -r(\partial_j a)r, \quad \partial_l \partial_j r = r(\partial_l a)r(\partial_j a)r + r(\partial_j a)r(\partial_l a)r - r(\partial_l \partial_j a)r. \quad (\text{MSI9})$$

These identities hold for matrices with no symmetry, normality, positivity, or commutation assumptions.

3. A cutoff inverse in the full mixed class

Theorem. Let $m \in \mathbb{R}$, $a \in S^{m,0}(\mathbb{R}^n \times \mathbb{R}^n; M_N(\mathbb{C}))$, and $\chi \in C_c^\infty(\mathbb{R}_\xi^n; \mathbb{C})$. Write

$$\theta(\xi) = 1 - \chi(\xi), \quad S = \text{supp } \theta. \quad (\text{MSI10})$$

Suppose that $a(x, \xi)$ is invertible whenever $x \in \mathbb{R}^n$ and $\xi \in S$, and that a constant $C_0 > 0$ satisfies

$$\|a(x, \xi)^{-1}\| \leq C_0(1 + |\xi|)^{-m} \quad (x \in \mathbb{R}^n, \xi \in S). \quad (\text{MSI11})$$

Define the function on the whole phase space by

$$b(x, \xi) = \begin{cases} \theta(\xi)a(x, \xi)^{-1}, & \xi \in S, \\ 0, & \xi \notin S. \end{cases} \quad (\text{MSI12})$$

Then $b \in S^{-m,0}$, and the two pointwise identities

$$a(x, \xi)b(x, \xi) = b(x, \xi)a(x, \xi) = \theta(\xi)I_N \quad (\text{MSI13})$$

hold on all of $\mathbb{R}^{2n}$. In particular $b = a^{-1}$ outside $\text{supp } \chi$. The theorem imposes no invertibility condition where θ is identically zero in a neighborhood, and imposes neither reality nor $0 \leq \chi \leq 1$ on the cutoff.

For $N = 1$, the scalar hypothesis

$$|a(x, \xi)| \geq c(1 + |\xi|)^m \quad (x \in \mathbb{R}^n, \xi \in S), \quad c > 0, \quad (\text{MSI14})$$

is sufficient, with $C_0 = c^{-1}$. In that case (MSI12) is precisely the smooth extension of $(1 - \chi(\xi))/a(x, \xi)$ by zero on the open region where $1 - \chi$ vanishes.

Proof, including seminorm constants. First we justify smoothness of the globally defined function. If $\xi_0 \notin S$, then θ is identically zero on a neighborhood of ξ_0 , so b is smooth near every (x_0, ξ_0) . If $\xi_0 \in S$, the invertible matrix $a(x_0, \xi_0)$ remains invertible on some open neighborhood of (x_0, ξ_0) , because its continuous determinant is nonzero there after shrinking the neighborhood. On that neighborhood the smooth function θa^{-1} agrees with (MSI12): at points outside S , $\theta = 0$. Thus it is a local smooth representative of b at every such point, including points of the boundary of S . This proves smoothness without imposing an additional neighborhood lower bound in the hypotheses. Formula (MSI13) follows directly on S ; outside S , both sides are zero.

For every combined multiindex $\nu = (\delta, \varepsilon)$, let

$$A_\nu = \|a\|_{m,0;\delta,\varepsilon}. \quad (\text{MSI15})$$

Define the nonnegative finite constants

$$Q_0 = C_0, \quad Q_\mu = \sum_{k=1}^{|\mu|} \sum_{\substack{\nu_1 + \dots + \nu_k = \mu \\ |\nu_i| > 0}} \frac{\mu!}{\nu_1! \dots \nu_k!} C_0^{k+1} \prod_{i=1}^k A_{\nu_i} \quad (\mu \neq 0). \quad (\text{MSI16})$$

At a point with $\xi \in S$, the inverse derivatives are understood through the local smooth inverse just constructed. Every term in (MSI6) has $k+1$ factors a^{-1} and k differentiated factors a . If $\nu_i = (\beta^{(i)}, \alpha^{(i)})$ and $\sum_i \nu_i = (\beta, \alpha)$, its total first-weight exponent is

$$-(k+1)m + \sum_{i=1}^k (m - \alpha_n^{(i)}) = -m - \alpha_n; \quad (\text{MSI17})$$

its second-weight exponent is $-\sum_i |\alpha'^{(i)}| = -|\alpha'|$. Therefore (MSI6), (MSI11), and the triangle inequality give the fully uniform bounds

$$\|\partial_x^\beta \partial_\xi^\alpha a^{-1}(x, \xi)\| \leq Q_{(\beta, \alpha)} W_{-m, 0; \alpha}(\xi) \quad (\xi \in S). \quad (\text{MSI18})$$

For $\alpha = \beta = 0$, this is exactly (MSI11), so no derivative induction lacks a starting case.

Choose a number $R \geq 1$ with $\text{supp } \chi \subset \{|\xi| \leq R\}$; $R = 1$ is allowed when $\chi = 0$. Set

$$T_\delta = \sup_\xi |\partial_\xi^\delta \theta(\xi)|, \quad L_\delta = \begin{cases} 1, & \delta = 0, \\ (1 + R)^{|\delta|}, & \delta \neq 0. \end{cases} \quad (\text{MSI19})$$

All T_δ are finite. For $\delta \neq 0$, the derivative $\partial^\delta \theta = -\partial^\delta \chi$ is supported in the ball of radius R . On that support,

$$\frac{W_{-m, 0; \alpha - \delta}(\xi)}{W_{-m, 0; \alpha}(\xi)} = (1 + |\xi|)^{\delta_n} (1 + |\xi'|)^{|\delta'|} \leq (1 + R)^{|\delta|} = L_\delta. \quad (\text{MSI20})$$

For $\delta = 0$, this ratio equals one everywhere. The product rule for $b = \theta a^{-1}$, valid locally at every point of S , now yields

$$\|b\|_{-m, 0; \beta, \alpha} \leq \sum_{\delta \leq \alpha} \binom{\alpha}{\delta} T_\delta L_\delta Q_{(\beta, \alpha - \delta)}. \quad (\text{MSI21})$$

To see why this is also a global estimate, at every point outside S all derivatives of b vanish on a neighborhood, and at every point of S the preceding local calculation applies. This includes every boundary point; no omitted limiting argument is needed. The right side of (MSI21) is a finite number depending only on the explicitly listed constants and derivative orders. Hence all seminorms required for $S^{-m, 0}$ are finite. In the scalar case, (MSI14) gives (MSI11) by taking reciprocals of the positive numerical lower bound. This proves every assertion. $\square$

4. Improvement for an individual frequency derivative

Theorem. Retain all hypotheses of Section 3. Fix one index $j \in \{1, \dots, n\}$. If

$$a_j := \partial_{\xi_j} a \in S^{m-1, 0}, \quad (\text{MSI22})$$

then

$$\partial_{\xi_j} b \in S^{-m-1, 0}. \quad (\text{MSI23})$$

There is no hypothesis here on the other first frequency derivatives beyond $a \in S^{m, 0}$. Consequently the same conclusion holds for any collection of indices for which (MSI22) holds, including all indices when the stronger assumption holds for all of them.

Proof. At every point with $\xi \in S$, the exact matrix identity (MSI9) gives

$$\partial_{\xi_j} b = -\theta a^{-1} a_j a^{-1} + (\partial_{\xi_j} \theta) a^{-1}. \quad (\text{MSI24})$$

The order of the two inverses and a_j is fixed. Each term has a smooth global extension by zero outside S . Indeed near a point of S it is the displayed product of smooth functions and the locally defined smooth inverse. At a point outside S , θ , all its derivatives, and the extended terms vanish on a neighborhood. On an overlap these local definitions agree. For the term involving $\partial_{\xi_j} \theta$, observe that every derivative of θ vanishes outside S , because θ is locally zero there. Thus the same gluing argument applies to that term.

We give explicit bounds for every derivative in (MSI24). Let

$$H_{\kappa}^{(j)} = \|a_j\|_{m-1,0;\kappa_x,\kappa_{\xi}} \quad \text{for } \kappa = (\kappa_x, \kappa_{\xi}) \in \mathbb{N}_0^{2n}. \quad (\text{MSI25})$$

When $\mu = (\beta, \alpha)$, a term in the repeated product rule for $\theta a^{-1} a_j a^{-1}$ is indexed by

$$(0, \delta) + \nu + \kappa + \lambda = \mu, \quad \delta \in \mathbb{N}_0^n, \quad \nu, \kappa, \lambda \in \mathbb{N}_0^{2n}. \quad (\text{MSI26})$$

The differentiated three matrix factors have combined first exponent $-m + (m-1) - m - (\alpha_n - \delta_n) = -m - 1 - \alpha_n + \delta_n$, and combined second exponent $-|\alpha'| + |\delta'|$. If $\delta = 0$, these are exactly the desired exponents. If $\delta \neq 0$, the cutoff derivative restricts the term to $|\xi| \leq R$, and the weight correction is bounded by L_{δ} , exactly as in (MSI20). Thus the (β, α) seminorm of the first term of (MSI24), in $S^{-m-1,0}$, is at most

$$U_{\mu}^{(j)} = \sum_{(0,\delta)+\nu+\kappa+\lambda=\mu} \frac{\mu!}{\delta! \nu! \kappa! \lambda!} T_{\delta} L_{\delta} Q_{\nu} H_{\kappa}^{(j)} Q_{\lambda}. \quad (\text{MSI27})$$

Here $(0, \delta)! = \delta!$, which accounts for the denominator. The finite multinomial count follows from the same derivative-choice argument that proved (MSI4), now with four factors.

For the second term of (MSI24), apply the product rule with $\delta \leq \alpha$ derivatives hitting $\partial_{\xi_j} \theta$. All such cutoff factors are supported in $|\xi| \leq R$, even for $\delta = 0$, because their total derivative order $|\delta| + 1$ is positive. Their weight comparison is

$$\frac{W_{-m,0;\alpha-\delta}(\xi)}{W_{-m-1,0;\alpha}(\xi)} = (1 + |\xi|)^{1+\delta_n} (1 + |\xi'|)^{|\delta'|} \leq (1 + R)^{1+|\delta|}. \quad (\text{MSI28})$$

If $e_j \in \mathbb{N}_0^n$ is the j -th frequency multiindex, the second term therefore has seminorm at most

$$V_{\mu}^{(j)} = \sum_{\delta \leq \alpha} \binom{\alpha}{\delta} T_{\delta+e_j} (1 + R)^{1+|\delta|} Q_{(\beta, \alpha-\delta)}. \quad (\text{MSI29})$$

Combining (MSI24), (MSI27), and (MSI29) gives

$$\|\partial_{\xi_j} b\|_{-m-1,0;\beta,\alpha} \leq U_{(\beta,\alpha)}^{(j)} + V_{(\beta,\alpha)}^{(j)} < \infty. \quad (\text{MSI30})$$

The local extension argument again makes these global bounds. This proves (MSI23) for the chosen j , without invoking any condition on another first derivative. $\square$

There is an immediate distinction between the normal and tangential indices. The normal case $j = n$ of (MSI22) follows already from $a \in S^{m,0}$, since

$$W_{m,0;\alpha+e_n} = W_{m-1,0;\alpha}. \quad (\text{MSI31})$$

For $j < n$, the original class instead yields $\partial_{\xi_j} a \in S^{m,-1}$. The first-frequency weight and second-frequency weight have different behavior when $|\xi_n| \rightarrow \infty$ while ξ' is fixed. The following explicit example proves that the tangential improvement cannot be inferred from the original class alone.

5. Two examples with different purposes

A singular matrix in the discarded region. Let $N = 2$, $n \geq 1$, $m = 2$, and

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad a(x, \xi) = |\xi|^2 I_2 + J. \quad (\text{MSI32})$$

This symbol is independent of x , and $J^2 = 0$, $\|J\| = 1$. It belongs to $S^{2,0}$, as may be checked at every derivative order directly. At order zero, $\|a\| \leq |\xi|^2 + 1 \leq (1 + |\xi|)^2$. At a first frequency derivative, $\partial_{\xi_j} a = 2\xi_j I_2$. For $j = n$, this is bounded by $2(1 + |\xi|)$, the required normal weight. For $j < n$, $|\xi_j| \leq |\xi'|$, so

$$2|\xi_j|(1 + |\xi'|) \leq 2|\xi'|^2(1 + |\xi'|) \leq 2(1 + |\xi|)^2. \quad (\text{MSI33})$$

At order two, the only nonzero derivatives are $\partial_{\xi_j}^2 a = 2I_2$. For $j = n$, the prescribed weight is one. For $j < n$, it is $(1 + |\xi|)^2(1 + |\xi'|)^{-2} \geq 1$. All mixed second derivatives, all derivatives of order greater than two, and every positive-order x derivative vanish. These observations prove all required seminorm bounds.

Every $a_j = 2\xi_j I_2$ also lies in $S^{1,0}$. At order zero its norm is at most $2(1 + |\xi|)$. Its only possibly nonzero positive-order derivative is $\partial_{\xi_j} a_j = 2I_2$; the required weight is one if $j = n$, and $(1 + |\xi|)/(1 + |\xi'|) \geq 1$ if $j < n$. All other derivatives vanish.

Choose a smooth cutoff $\chi(\xi)$ equal to one when $|\xi| \leq 1$, equal to zero when $|\xi| \geq 2$, and taking values in $[0, 1]$. Such a cutoff can be made explicitly. Define $h(t) = e^{-1/t}$ for $t > 0$ and $h(t) = 0$ for $t \leq 0$; repeated differentiation for $t > 0$ gives a polynomial in t^{-1} times $e^{-1/t}$, which tends to zero as $t \downarrow 0$, at every derivative order. Thus h is smooth. Set

$$\chi(\xi) = \frac{h(4 - |\xi|^2)}{h(4 - |\xi|^2) + h(|\xi|^2 - 1)}. \quad (\text{MSI34})$$

The denominator is strictly positive for every ξ : if $|\xi|^2 \leq 1$, the first term is positive; if $|\xi|^2 \geq 4$, the second is positive; in between both are positive. This proves smoothness and all stated cutoff properties, including compact support. Every $\xi \in \text{supp}(1 - \chi)$ has $|\xi| \geq 1$. For such ξ , direct multiplication using $J^2 = 0$ gives

$$a(x, \xi)^{-1} = |\xi|^{-2} I_2 - |\xi|^{-4} J, \quad \|a(x, \xi)^{-1}\| \leq 2|\xi|^{-2} \leq 8(1 + |\xi|)^{-2}. \quad (\text{MSI35})$$

Thus (MSI11) holds with $C_0 = 8$. The resulting b belongs to $S^{-2,0}$, and every frequency derivative belongs to $S^{-3,0}$. Although $a(x, 0) = J$ is singular, $b = 0$ on $|\xi| \leq 1$, and the globally smooth inverse construction remains valid. This verifies concretely why global invertibility of a is unnecessary.

A tangential derivative that does not improve. Let $n \geq 2$, and choose a real-valued $\varphi \in C_c^\infty(\mathbb{R}^{n-1})$ with $|\varphi| \leq 1/2$ and $\partial_{\xi_1} \varphi(\eta) \neq 0$ at some point η . Such a function is obtained by multiplying the coordinate ξ_1 by a smooth cutoff equal to one near the origin and then multiplying by a sufficiently small positive constant; the cutoff is constructed by the same one-variable function h as in (MSI34). Put

$$a(x, \xi) = 2 + \varphi(\xi'), \quad \chi = 0, \quad b(x, \xi) = \frac{1}{2 + \varphi(\xi')}. \quad (\text{MSI36})$$

The symbol a is scalar and at least $3/2$. To verify $a \in S^{0,0}$, if $\alpha_n > 0$ or $\beta \neq 0$, the required derivative is zero. If $\beta = 0$, $\alpha_n = 0$, and $|\alpha'| > 0$, the derivative $\partial_{\xi'}^{\alpha'} \varphi$ has compact support in ξ' . If that support lies in $|\xi'| \leq R_\varphi$, the required seminorm is at most $\|\partial^{\alpha'} \varphi\|_\infty (1 + R_\varphi)^{|\alpha'|}$. The order-zero seminorm is at most $5/2$. Thus every required estimate is established. The inverse bound (MSI11) holds for $m = 0$ with $C_0 = 2/3$, and Section 3 yields $b \in S^{0,0}$.

At $\xi' = \eta$, however,

$$\partial_{\xi_1} b(x, \eta, \xi_n) = -\frac{\partial_{\xi_1} \varphi(\eta)}{(2 + \varphi(\eta))^2} \neq 0, \quad (\text{MSI37})$$

independently of ξ_n . The order-zero estimate for membership in $S^{-1,0}$ would bound the absolute value of this nonzero constant by $C(1 + \sqrt{|\eta|^2 + \xi_n^2})^{-1}$ for all ξ_n . The right side tends to zero as $|\xi_n| \rightarrow \infty$, which is impossible. Therefore $\partial_{\xi_1} b \notin S^{-1,0}$. The same argument shows $\partial_{\xi_1} a \notin S^{-1,0}$. This example preserves the full original mixed class while proving that the extra tangential assumption in (MSI22) has mathematical content.

6. Solved exercise: detect an invalid commutation

Exercise. For $n \geq 1$, take a matrix symbol independent of ξ and of $x_2, \dots, x_n$, defined by

$$a(x, \xi) = \begin{pmatrix} 1 & \sin x_1 \\ \cos x_1 & 1 \end{pmatrix}, \quad \chi = 0. \quad (\text{MSI38})$$

Prove that Section 3 applies with $m = 0$. Compute b , $\partial_{x_1} b$, and $\partial_{x_1}^2 b$ at $x_1 = 0$. Determine whether replacing the inverse derivative by $-(\partial_{x_1} a)b^2$ gives the correct first derivative there. Finally, explain the frequency-derivative conclusion for this example.

Solution. Every x_1 derivative of every entry of a is bounded, and all positive-order frequency derivatives vanish. These facts verify every seminorm of $S^{0,0}$: when $\alpha = 0$, the denominator in (MSI3) is one; when $\alpha \neq 0$, the numerator vanishes. The determinant is

$$d(x_1) = 1 - \sin x_1 \cos x_1 \geq \frac{1}{2}, \quad b(x, \xi) = \frac{1}{d(x_1)} \begin{pmatrix} 1 & -\sin x_1 \\ -\cos x_1 & 1 \end{pmatrix}. \quad (\text{MSI39})$$

The bound on the determinant follows from $2|\sin x_1 \cos x_1| \leq \sin^2 x_1 + \cos^2 x_1 = 1$. The numerator matrix has squared Frobenius norm $2 + \sin^2 x_1 + \cos^2 x_1 = 3$, so its Euclidean operator norm is at most $\sqrt{3}$: for any vector v , the scalar Cauchy–Schwarz inequality applied to each row gives $\|Mv\|^2 \leq (\sum_{k,l} |M_{kl}|^2) \|v\|^2$.

Consequently $\|b\| \leq 2\sqrt{3}$, and (MSI11) holds with $C_0 = 2\sqrt{3}$. In particular the theorem supplies all inverse-symbol seminorms without assuming them in advance.

At $x_1 = 0$, write

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad R = A^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}. \quad (\text{MSI40})$$

Formula (MSI9), with the products performed in their written order, gives

$$\partial_{x_1} b|_{x_1=0} = -RA_1R = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (\text{MSI41})$$

and

$$\partial_{x_1}^2 b|_{x_1=0} = 2RA_1RA_1R - RA_2R = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}. \quad (\text{MSI42})$$

For a direct check of the first multiplication, $RA_1R = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. For the second, $RA_1RA_1R = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $RA_2R = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}$; these two explicit products yield (MSI42).

By contrast,

$$-A_1R^2 = \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (\text{MSI43})$$

The proposed commutation therefore fails even for a uniformly invertible, smooth, globally bounded symbol. Every positive-order frequency derivative of a and b is zero. The symbols themselves have bounded position derivatives of every order, so both belong to $S^{0,0}$. Indeed their entries are rational functions of sine and cosine with denominators bounded away from zero; repeated position differentiation therefore remains bounded. The nonzero symbols are not asserted to belong to every mixed order class. In particular (MSI22) and (MSI23) hold for each $j \in \{1, \dots, n\}$ with $m = 0$. This last conclusion comes from actual vanishing, whereas the noncommuting x_1 derivatives above remain nonzero. $\square$

References